Cycle-and-period design sweep: planning note

2026-05-07 06:47 PDT

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Purpose. This document specifies the parameter grid, simulation plan, and analysis approach for a sensitivity sweep that quantifies how the biomarker-treatment interaction test ($H_0 : \beta_{bm:D} = 0$) responds to changes in the protocol-level design parameters that govern within-participant treatment cycling: the number of on/off cycles, the on-drug and off-drug period lengths within each cycle, the symmetry of the on/off split, and the optional run-in and run-out phases. It is a plan; it reports no results.

The sweep is intended to inform protocol-design decisions for the prazosin-PTSD aggregated N-of-1 application that motivates the pmsimstats project [1]. It is the **S13** sensitivity block of paper 05 (cited as 05-S13 across manuscripts); numbering continues from the existing S1-S12 blocks, whose drivers are `analysis/scripts/nof1-design-sensitivity/01-effect-size.R` to `12-morris-null.R`.

Status note, 2026-10-07. Tier 1 has been implemented as `13-cycle-period.R` in `analysis/scripts/nof1-design-sensitivity/`, with `build_cycle_design()` in `00-common.R`. The driver departs from this plan in one axis: it uses five values of c_{bm} (0, 0.10, 0.20, 0.30, 0.45) rather than two. The stored output `sensitivity-13-cycle-period.rds` in `analysis/data/derived_data/sensitivity/` holds 40,500 fits, which is the 810 Tier 1 cells at 50 replicates each (dev scale), not the 500-replicate production run (inspected). Tiers 2a to 2d were not found in the scripts.

Contents

Notation index and glossary 4
1. Scientific motivation 5
2. Design parameterization 6
    2a. Terminology 6
    2b. Sampling density convention 7
    2c. Helper 7
3. Sweep design 8
    3a. Tier 1: principal $k \times T_{\text{on}} \times T_{\text{off}}$ factorial 8
    3b. Tier 2a: within-cycle asymmetry sweep 9
    3c. Tier 2b: across-cycle asymmetry sweep 10
    3d. Tier 2c: sampling-density sweep 10
    3e. Tier 2d: $N \times k$ trade-off 11
    Sweep totals 11
4. Theoretical expectations 12
5. Performance measures 13
6. Visualization 13
7. Implementation plan 14
8. Risks and what they would mean 14
9. Reporting hooks for the manuscript 15
10. Resolved decisions 15
References 16

List of Tables

1 Notation index and glossary of symbols used in this document . . 4
2 Tier 1 principal factorial axes over cycles and period lengths . . . 8
3 Tier 2a within-cycle asymmetry sweep axes at $k = 2$ 9
4 Tier 2b across-cycle asymmetry sweep axes at fixed 16-week duration 10
5 Tier 2c sampling-density sweep axes at the reference cell 11

6 Tier 2d sweep axes for the N by k trade-off 11

7 Cells, replicates and cell-replicates for each tier of the S13 sweep (Tier 1 corrected from 264 to 324 cells on 2026-10-07) 11

Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
$\beta_{bm:D}$	biomarker-by-treatment interaction coefficient (`bm:Dbc`)	canonical
D_{it}, $D_{bc,it}$	binary drug state (code `Db`); exposure-decayed drug indicator (code `Dbc`)	canonical
B_i	biomarker	canonical
k	number of blinded on/off cycles	canonical
T_{on}, T_{off}	on-drug and off-drug period lengths within a cycle (weeks)	local
T_1, T_2	lengths of cycles 1 and 2 (Tier 2b)	local
T_{runin}, $T_{\text{openlabel}}$, T_{runout}	run-in, open-label and run-out durations (weeks)	local
N	total sample size across randomization paths	canonical
t, $t_{1/2}$	time in weeks; carryover half-life (weeks)	canonical
c_{bm}	biomarker moderation strength; a shared label across the two architectures (see glossary)	canonical
σ_{BR}, θ_{true}	BR standard deviation; calibrated true value $-c_{bm}\sigma_{BR}$	canonical
ρ	AR(1) within-factor serial correlation	canonical
α	nominal significance level	canonical
$I(\cdot)$	Fisher information	local

Symbol	Meaning	Status
σ^2	residual variance of a weekly observation in the Section 4 approximation	local
b_i, s_b^2	standardized biomarker; its sample variance across participants	canonical; local
η_t	expectancy weight at visit t (1 open label, 0.5 blinded)	canonical
n_{on}, n_{off}	numbers of on-drug and off-drug weekly observations per participant (2026-10-08 corrections)	local

- **Shared c_{bm} label.** Tables cross both architectures, so c_{bm} labels the moderation strength at a matched value: the correlation c_{bm} under covariance moderation and the multiplier β_{bm} under mean moderation (NOTATION.md, rule 2).
- **Covariance moderation, mean moderation.** The two data-generating architectures (formerly Architecture B and A).
- **E1, E2, E3.** Binary on-drug indicator; exposure-decayed $D_{bc,it}$; binary plus a lagged off-drug term (formerly labeled `A1`, `A2`, `A3`).
- **Run-in, open-label.** Defined in Section 2a.

1. Scientific motivation

Existing pmsimstats sweeps (S1-S12) examine sensitivity of the **main drug effect** test under axis-wise perturbations of single parameters around a reference design (the Hendrickson Hybrid). These sweeps hold the protocol structure fixed and vary the data-generating process. The complementary question, which the existing infrastructure does not answer, is how the interaction test responds to changes in the **protocol structure itself**:

1. **Number of cycles k.** More within-participant treatment switches contribute more within-participant contrasts to the estimator of $\beta_{bm:D}$. Under the simplest balanced design with no carryover, the Fisher information on the interaction coefficient is approximately linear in k. Whether this scaling holds at realistic prazosin-PTSD parameters with carryover is the empirical question.

2. **Period lengths** $T_{\text{on}}, T_{\text{off}}$. The on-drug period must be long enough for the BR Gompertz trajectory to develop measurable signal; the off-drug period must be long enough for residual drug effect to wash out and restore the on-drug-versus-off-drug contrast. The two requirements compete at fixed total trial duration.
3. **Within-cycle asymmetry** $T_{\text{on}} \neq T_{\text{off}}$. Pharmacokinetic onset and washout typically operate on different timescales; the interaction-information harmonic mean argument (Section 4 below) predicts that symmetric periods are optimal at fixed total duration absent carryover, but carryover should bend the optimum toward longer off-drug periods. (Corrected 2026-10-08: the symmetric optimum holds only without open-label, run-in or run-out weeks, or with a biomarker-by-expectancy term in the analysis. Under the planned E2 analysis with 8 open-label weeks the no-carryover optimum of the Tier 2a sweep is $T_{\text{on}} = 2$; see Section 4.)
4. **Across-cycle asymmetry** ($T_1/T_2 \neq 1$). Cycle 1 may reasonably differ in length from later cycles, particularly if it doubles as titration. The empirical question is whether such asymmetry materially affects power.
5. **Run-in and run-out phases**. Open-label titration before blinding (the OL+BDC pattern) and post-cycle observation contribute information differently from within-cycle observations. Including these as design parameters allows the sweep to characterize their power contribution.

A reviewer of any pmsimstats manuscript can reasonably ask, 'if the trialist reduces the number of cycles to fit within a 12-week window, by how much does the interaction power decrease?' The S13 sweep is intended to answer such questions quantitatively.

2. Design parameterization

2a. Terminology

We distinguish two concepts that are conflated in much of the clinical-trial methodology literature:

- **Run-in.** A period before treatment starts. Observations during run-in are pre-treatment baseline measurements. Drug is *not* given. Run-in observations have $D_{it} = 0$ and $D_{bc,it} = 0$ in the analysis model.
- **Open-label.** A period during which active drug is given and the participant is aware that the treatment is active. Open-label observations have $D_{it} = 1$ and $D_{bc,it} = 1$ in the analysis model.

Run-in and open-label are orthogonal: a trial may have either, both, or neither. The Hendrickson 2020 designs have open-label phases (OL1, OL2 at weeks 4, 8

in the Hybrid; OL1-OL4 at weeks 4-16 in the OL+BDC) but no run-in: every patient receives active drug from week 0 onward.

The two phases also play different roles in identifying the interaction coefficient $\beta_{bm:D}$ (argued):

- Run-in observations contribute *directly* to identifying the interaction, because they are off-drug ($D_{bc,it} = 0$) observations with biomarker variation, contrasting against the later on-drug observations.
- Open-label observations contribute *indirectly* (through random-intercept precision, AR(1) parameter estimation, and linear-in-t slope absorption) but not directly: their $D_{bc,it} = 1$ value is identical to the within-cycle on-drug observations, so they do not enlarge the on/off contrast.

(Corrected 2026-10-08: the open-label statement holds only for an analysis that includes a biomarker-by-expectancy (or biomarker-by-phase) term. Under the planned E2 analysis, which has none, the open-label weeks are on-drug observations of the on/off contrast and enter it directly: the on-drug count is $n_{\text{on}} = T_{\text{openlabel}} + kT_{\text{on}}$ in $\text{Var}(\widehat{\beta}_{bm:D}) \propto (n_{\text{on}} + n_{\text{off}})/(n_{\text{on}} n_{\text{off}})$ (derived; verified from the GLS information of the full design matrix, `analysis/scripts/compendium-mathematics/check-gompertz-rmanova-cycles.R`). Because the open-label visits carry $\eta_t = 1$, they are also the visits through which expectancy confounding enters under a PB-coupled biomarker. See Section 4 and `analysis/report/compendium-mathematics.tex`, Section "Cycle count and period length" (`sec:invb-cycles`).)

The S13 sweep treats the two as separate parameters.

2b. Sampling density convention

We standardize on **one observation per week** throughout all phases as the default sampling density for S13. This is dense enough to track the Gompertz BR trajectory in the prazosin-PTSD parameter regime, sparse enough to avoid near-singularity in the AR(1) covariance when ρ is large, and matches the within-cycle density used in the Hendrickson Hybrid blinded discontinuation phase. Under this default, the number of observations per phase is directly proportional to the phase duration: a 4-week on-drug period has 4 observations, a 2-week off-drug period has 2 observations. A separate Tier 2 block (Section 3d below) varies the density to characterize the sampling-cost frontier.

2c. Helper

The current `buildtrialdesign()` interface takes a binary `ondrug` vector at user-specified `timepoints`. This is expressive (any sequence is encodable) but tedious for protocol-level sweeps. We introduce a higher-level helper:

```
build_cycle_design(
  T_runin     = 0,      # weeks of pre-treatment baseline (no drug)
  T_openlabel = 0,      # weeks of on-drug, unblinded titration
  k,                    # number of blinded on/off cycles
  T_on,                 # blinded on-drug period (weeks); scalar
                        #   or length-k vector
  T_off,                # blinded off-drug period; scalar or length-k
  T_runout    = 0,      # post-cycle observation weeks
  order       = 'on_first',  # 'on_first' or 'off_first'
  density     = 1,      # observations per week (default 1/wk)
  expectancy  = 0.5     # blinded-phase expectancy
)
```

`build_cycle_design()` constructs the timepoints, on-drug binary vector, and expectancy vector, then delegates to `buildtrialdesign()`. The downstream simulation core (`generate_data`, `lme_analysis`, `prepare_long_data`) consumes the result unchanged.

A round-trip equivalence check is part of the implementation: calling `build_cycle_design(T_openlabel = 8, k = 1, T_on = 0, T_off = 4, T_runout = 8, density = 1)` with appropriate density adjustment must reproduce the canonical Hendrickson Hybrid schedule (open-label observations at weeks 4 and 8; blinded discontinuation observations at weeks 9, 10, 11, 12; brief crossover at weeks 16 and 20). The Hendrickson schedule uses phase-specific densities (1/4 wk for open-label, 1/wk for blinded discontinuation, 1/6 wk for brief crossover); a separate helper `build_hendrickson_design()` will be provided for exact Hendrickson reproduction.

3. Sweep design

3a. Tier 1: principal $k \times T_{\text{on}} \times T_{\text{off}}$ factorial

Symmetric across-cycle ($T_1/T_2 = 1$), focused on the founding interaction question. Sampling density held at 1 observation per week throughout all phases (so number of observations per phase equals the phase duration in weeks); run-in held at 0 weeks (no pre-treatment baseline, matching Hendrickson); open-label held at 8 weeks (matching the Hendrickson Hybrid two-observation titration).

Table 2: Tier 1 principal factorial axes over cycles and period lengths

Axis	Values	Levels
Cycles k	1, 2, 3, 4	4
On-drug period T_{on} (weeks)	2, 4, 6	3
Off-drug period T_{off} (weeks)	2, 4, 6	3
DGP architecture	mean moderation, covariance moderation	2

Axis	Values	Levels
Carryover half-life $t_{1/2}$ (weeks)	0, 0.5, 1.0	3
Biomarker moderation c_{bm} (shared label)	0, 0.45	2

Held fixed: $T_{\text{runin}} = 0$, $T_{\text{openlabel}} = 8$, $T_{\text{runout}} = 0$, density = 1 / wk, $N = 70$.

Cell count: $4 \times 3 \times 3 \times 2 \times 3 \times 2 = 432$ combinations. **Feasibility filter.** Total time per participant is $T_{\text{runin}} + T_{\text{openlabel}} + k \cdot (T_{\text{on}} + T_{\text{off}}) + T_{\text{runout}}$. With the held-fixed values above, the constraint 'total trial duration ≤ 32 weeks', that is $k(T_{\text{on}} + T_{\text{off}}) \leq 24$, eliminates the longest-duration cells ($k = 4$ at $T_{\text{on}} = T_{\text{off}} = 6$ is $8 + 4 \times 12 = 56$ weeks; $k = 3$ at $T_{\text{on}} = T_{\text{off}} = 6$ is 44 weeks). It keeps all 9 $(T_{\text{on}}, T_{\text{off}})$ pairs at $k = 1$ and $k = 2$, 6 at $k = 3$ and 3 at $k = 4$: 27 of the 36 cycle structures, hence $27 \times 12 = 324$ cells (derived; the 27 structures match the design grid stored by `13-cycle-period.R`, verified 2026-10-07). An earlier version of this paragraph gave about 264 cells. The 32-week cap is a planning assumption about patient feasibility for N-of-1 trials in chronic neurobehavioral conditions, where adherence and dropout may become limiting at long durations. (The pharmacotherapy meta-analysis previously cited here [2] does not address trial-duration feasibility, so the cap rests on judgment, not on that source.) The 324 cells constitute Tier 1 of the sweep.

Reference cell, used as the anchor for asymmetric extensions in Tier 2: $k = 2$, $T_{\text{on}} = T_{\text{off}} = 4$, covariance moderation, $t_{1/2} = 0.5$, $c_{bm} = 0.45$, $N = 70$, $T_{\text{openlabel}} = 8$. This is approximately the Hendrickson-2020 Hybrid configuration, expressed in cycle-period parameters with the standardized 1-per-week sampling density extended across the cycles (where the published Hybrid uses 1 / wk during blinded discontinuation but coarser density during open-label and brief crossover).

Sample size N is fixed at 70 for Tier 1 to keep cell count manageable; an $N = 35$ sweep is added in Tier 2.

3b. Tier 2a: within-cycle asymmetry sweep

At fixed $k = 2$ and matched total period length $T_{\text{on}} + T_{\text{off}} = 8$ weeks, walk the asymmetry ratio.

Table 3: Tier 2a within-cycle asymmetry sweep axes at $k = 2$

Axis	Values	Levels
$(T_{\text{on}}, T_{\text{off}})$	(2,6), (3,5), (4,4), (5,3), (6,2)	5
Architecture	mean, covariance moderation	2
$t_{1/2}$	0, 0.5, 1.0	3

Axis	Values	Levels
c_{bm}	0, 0.45	2

Cell count: $5 \times 2 \times 3 \times 2 = 60$. Sample size fixed at $N = 70$. The expected pattern under the harmonic-mean prediction (Section 4) is a maximum at $(4, 4)$ and symmetric decay; the expected pattern under carryover is a minimum at $(6, 2)$ (insufficient washout) and a non-symmetric maximum biased toward longer off-drug periods. (Corrected 2026-10-08: under the planned E2 analysis model, which has no biomarker-by-expectancy term, the eight open-label weeks count as on-drug observations, and the no-carryover information per unit $(N-1)s_b^2/\sigma^2$ is 6.00, 5.83, 5.33, 4.50 and 3.33 at $T_{\text{on}} = 2, \ldots, 6$: the predicted maximum is at $(2, 6)$, not $(4, 4)$, and the decay is not symmetric. The symmetric pattern, 3.00, 3.75, 4.00, 3.75 and 3.00, applies only if a $b_i\eta_t$ term is added (verified, `analysis/scripts/compendium-mathematics/check-gompertz-rmanova-cycles.R`; see Section 4).)

3c. Tier 2b: across-cycle asymmetry sweep

At fixed $k = 2$, $T_{\text{on}} = T_{\text{off}}$, walk the relative period length of cycle 1 vs cycle 2 keeping total duration constant at 16 weeks.

Table 4: Tier 2b across-cycle asymmetry sweep axes at fixed 16-week duration

Axis	Values	Levels
Cycle-length ratio T_1/T_2	0.5, 0.75, 1.0, 1.5, 2.0	5
Architecture	mean, covariance moderation	2
$t_{1/2}$	0, 0.5, 1.0	3
c_{bm}	0.45	1

Cell count: $5 \times 2 \times 3 \times 1 = 30$.

3d. Tier 2c: sampling-density sweep

Vary the standardized sampling density (in observations per week) at the reference $(k, T_{\text{on}}, T_{\text{off}})$ to characterize the sampling-cost frontier and the AR(1)-bounded marginal value of additional observations. The principal sweep runs at 1/wk; this Tier 2c block extends below and above that default.

Table 5: Tier 2c sampling-density sweep axes at the reference cell

Axis	Values	Levels
Density (observations per week)	0.5, 1, 2, 4	4
Architecture	mean, covariance moderation	2
$t_{1/2}$	0, 0.5, 1.0	3
c_{bm}	0.45	1

Cell count: $4 \times 2 \times 3 \times 1 = 24$. The 0.5 / wk level halves the observation count and reflects monthly visits; the 4 / wk level quadruples it and reflects daily symptom self-reporting. Held fixed at the Tier 1 reference cell otherwise.

3e. Tier 2d: $N \times k$ trade-off

A two-axis sweep showing whether more cycles per patient can substitute for more patients at fixed total person-weeks. This is the design-economics question for trialists choosing between recruiting an additional participant and adding a cycle to the existing protocol.

Table 6: Tier 2d sweep axes for the N by k trade-off

Axis	Values	Levels
N	20, 35, 50, 70	4
k	1, 2, 3	3
Architecture	covariance moderation	1
$t_{1/2}$	0.5	1
c_{bm}	0.45	1

Cell count: $4 \times 3 = 12$.

Sweep totals

Table 7: Cells, replicates and cell-replicates for each tier of the S13 sweep (Tier 1 corrected from 264 to 324 cells on 2026-10-07)

Tier	Cells	Replicates	Cell-replicates
1: principal $k \times T_{\text{on}} \times T_{\text{off}}$	324	500	162,000
2a: within-cycle asymmetry	60	500	30,000

Tier	Cells	Replicates	Cell-replicates
2b: across-cycle asymmetry	30	500	15,000
2c: sampling density	24	500	12,000
2d: $N \times k$ trade-off	12	500	6,000
Total	450		225,000

At the Tier 1 production-run rate (approximately 13 cell-replicates per second on the existing 8-core hardware, per the carryover-sensitivity production timing; not re-measured), the full sweep completes in approximately 4.8 hours. Dev-scale (50 replicates per cell) finishes in approximately 29 minutes (derived from the stated rate).

4. Theoretical expectations

Under the mean-moderation architecture with no carryover, no AR(1) correlation, and balanced symmetric cycles, the Fisher information on $\beta_{bm:D}$ scales approximately as

$$I(\beta_{bm:D}) \;\propto\; N \cdot k \cdot \frac{T_{\text{on}} T_{\text{off}}}{T_{\text{on}} + T_{\text{off}}} \cdot \text{Var}(B_i)$$

at leading order (derived, at one observation per week). The harmonic-mean term arises from the within-participant contrast: with independent errors of variance σ^2, an on-minus-off mean difference over T_{on} and T_{off} weekly observations has variance $\sigma^2(T_{\text{on}} + T_{\text{off}})/(T_{\text{on}} T_{\text{off}})$. With no carryover, $D_{bc,it}$ equals the binary D_{it}. Three predictions follow:

1. **Linear in k.** Doubling k at fixed $(T_{\text{on}}, T_{\text{off}})$ approximately doubles information, so the $\hat{\beta}_{bm:D}$ standard error shrinks by a factor $\sqrt{2}$.
2. **Symmetric optimum.** At fixed total period length $T_{\text{on}} + T_{\text{off}}$, the harmonic-mean term is maximized at $T_{\text{on}} = T_{\text{off}}$.
3. **Carryover bends the optimum.** When residual drug effect contaminates the early off-drug period, the effective on/off contrast in $D_{bc,it}$ is reduced (argued). The optimal asymmetry is then $T_{\text{off}} > T_{\text{on}}$ to recover usable washout.

(Corrected 2026-10-08: the formula above and predictions 1 and 2 hold only when the design has no open-label, run-in or run-out weeks, or when the analysis includes a biomarker-by-expectancy term $b_i \eta_t$. With independent errors and a centered biomarker, the exact result is

$\mathrm{Var}(\hat{\beta}_{bm:D}) = \sigma^2(n_{\text{on}} + n_{\text{off}})/[(N-1)s_b^2\, n_{\text{on}} n_{\text{off}}]$, with $n_{\text{on}} = T_{\text{openlabel}} + kT_{\text{on}}$ and $n_{\text{off}} = T_{\text{runin}} + kT_{\text{off}} + T_{\text{runout}}$; a linear time term or an expectancy main effect leaves it unchanged, and a $b_i\eta_t$ term removes the open-label weeks from n_{on} (derived; verified to machine precision for every Tier 1 cycle structure). Under the planned E2 analysis, with $T_{\text{openlabel}} = 8$, linearity in k fails (1.67, 3.00, 4.20 and 5.33 for $k = 1$ to 4 at $T_{\text{on}} = T_{\text{off}} = 2$, per unit $(N-1)s_b^2/\sigma^2$), and at fixed $S = T_{\text{on}} + T_{\text{off}}$ the optimum is $T_{\text{on}}^* = S/2 - (T_{\text{openlabel}} - T_{\text{runin}} - T_{\text{runout}})/(2k)$, truncated to $[0, S]$. For Tier 2a ($k = 2$, $S = 8$) this is $T_{\text{on}}^* = 2$, not the symmetric $(4, 4)$. Prediction 3 is unaffected as an argument but is now relative to this asymmetric baseline. Verified, `analysis/scripts/compendium-mathematics/check-gompertz-rmanova-cycles.R`; see `analysis/report/compendium-mathematics.tex`, Section "Cycle count and period length" (`sec:invb-cycles`).)

The empirical task is to confirm or contradict each prediction under the Gompertz BR trajectory and AR(1) correlation that pmsimstats actually implements.

5. Performance measures

For each cell, report the Morris-style metrics [3, Table 6] now standard in pmsimstats:

1. Power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, with Monte Carlo SE.
2. Coverage of nominal 95 percent Wald CI, with MC SE.
3. Bias on the calibrated effect-size scale ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$), with MC SE.
4. Empirical SE.
5. Mean squared error.
6. Non-convergence rate per cell.

6. Visualization

- **Tier 1 heatmap matrix.** T_{on} on x, T_{off} on y, one row of panels per k and one column per carryover half-life (three). Color encodes power.
- **Cycles-per-patient line plot.** Power versus k at fixed $T_{\text{on}} = T_{\text{off}} = 4$, faceted by carryover.
- **Asymmetry curve.** Power versus $T_{\text{on}}/T_{\text{off}}$ ratio at fixed total period length, faceted by carryover.
- **N-versus-k trade-off plot.** Power as a function of total person-weeks ($N \cdot k \cdot (T_{\text{on}} + T_{\text{off}})$), with (N, k) combinations labeled. The question is whether the points fall on a single curve (perfect trade-off) or whether k dominates N or vice versa.
- **Lasagna plot of all 450 cells.** Same convention as `analysis/figures/02-lasagna-power.pdf`.

7. Implementation plan

1. **Helper.** `build_cycle_design()` in `analysis/scripts/nof1-design-sensitivity/simulation-core` (or a shared file `analysis/scripts/_shared/design-helpers.R` if multiple sweeps will reuse it). Include unit tests that round-trip against the existing OL, CO, Hybrid, and OL+BDC presets.
2. **Driver.** New file `analysis/scripts/nof1-design-sensitivity/13-cycle-period.R` following the pattern of the existing 01-12 sensitivity-block drivers. Writes `analysis/scripts/nof1-design-sensitivity/output/13-cycle-period.rds`.
3. **Summarizer.** Extend `analysis/scripts/nof1-design-sensitivity/summarise.R` (or the carryover-sensitivity equivalent) to handle the new block. Same Morris-compliant output columns as the existing summarizer. (No file of that name exists as of 2026-10-07; the directory holds `run-all.R` and `plot-all.R` instead.)
4. **Figures.** New file `analysis/scripts/nof1-design-sensitivity/13-cycle-period-figures.R` producing the heatmap matrix, line plot, asymmetry curve, trade-off plot, and lasagna plot.
5. **Manuscript prose.** Add §S13 to `analysis/report/05-nof1-design-sensitivity/report.Rmd` with abstract revision to mention the design-parameter sweep, a new Methods subsection, and a Results subsection per the figure set.
6. **Reproducibility.** Production run records master seed, commit SHA, R version, package versions in the output metadata, per the conventions in `docs/feedback_morris_simulation_standards.md`. (That file was not found in the repository on 2026-10-07.)

8. Risks and what they would mean

- **Linear-in-k prediction fails.** If doubling k does not approximately double information, the Fisher-information derivation in Section 4 is missing a term. Most likely candidate: AR(1) correlation reduces the effective independence of cycle-level contrasts. Diagnostic: rerun the Tier 1 grid at $\rho = 0$ (no within-factor serial correlation, so that with the random intercept the total covariance is compound-symmetric) to isolate the AR(1) contribution.
- **Symmetric-optimum prediction fails even without carryover.** If the asymmetry sweep at $t_{1/2} = 0$ does not peak at $(4, 4)$, the Gompertz curvature is contributing a non- negligible asymmetric term to the Fisher information. This would be a material finding that overturns the textbook recommendation for symmetric crossovers. (Corrected 2026-10-08: under the planned E2 analysis with eight open-label weeks, a peak at the shortest on-drug period, $(2, 6)$, is the expected no-carryover result and does not by itself indicate a Gompertz curvature term; see Section 4.)
- **Carryover does not bend the optimum.** If the asymmetry sweep at $t_{1/2} > 0$ also peaks near symmetric, the carryover effect on the optimum design is smaller than the direct effect on power. This would simplify trial-

design recommendations: symmetric is good across the realistic carryover range.

- **Across-cycle asymmetry has near-zero effect.** If the Tier 2b sweep is essentially flat, then the choice of cycle-1 versus cycle-2 length is not load-bearing for the interaction test. This would be a simplifying finding and would justify treating T_{on} and T_{off} as scalars throughout the manuscript.
- **N-versus-k are not interchangeable.** If the Tier 2d trade-off plot shows that N dominates k (or vice versa) at fixed total person-weeks, the design-economics conclusion is unambiguous. If the curves coincide, the trialist is free to choose based on practical considerations (recruitment cost, adherence).

9. Reporting hooks for the manuscript

The S13 results should support three direct statements in the nof1-design-sensitivity manuscript:

1. A trade-off curve quantifying how much per-patient duration ($k \cdot (T_{\text{on}} + T_{\text{off}})$) the trial must accept to reach 80 percent power at the prazosin-PTSD design parameters.
2. A recommendation on cycle count: at the prazosin-PTSD reference parameters, the marginal value of adding the third cycle is X percentage points of power, the fourth cycle is Y, and the recommended k is whichever delivers the target power most economically.
3. A statement on asymmetry: whether the textbook symmetric recommendation is robust to carryover, or whether a specific asymmetric form is preferred for the prazosin-PTSD biology.

10. Resolved decisions

- Open-label duration as a sweep axis: **resolved (2026-05-07).** Held fixed at $T_{\text{openlabel}} = 8$ weeks, matching the Hendrickson Hybrid two-observation titration. Not swept in S13.
- True run-in (pre-treatment baseline) as a sweep axis: **resolved (2026-05-07).** Held fixed at $T_{\text{runin}} = 0$ weeks, matching the Hendrickson convention. Not swept in S13. Note that run-in observations contribute *directly* to identifying the interaction coefficient (clean off-drug observations unaffected by carryover); a future companion sweep could quantify that contribution if the question becomes relevant for protocol design.
- Analysis-specification axis: **resolved (2026-05-07).** Held fixed at E2 (matched-decay $D_{bc,it}$) for Tier 1, on the stated ground that E2 dominates E1 and E3 across the parameter space. *Note, 2026-10-07:* that ground does not hold in the stored carryover-sensitivity grid (`analysis/data/02-grid-summary.rds`, $N = 70$). E2 is nearly unbiased

under covariance moderation, where E1 and E3 attenuate by up to 42 percent at $t_{1/2} = 1$, but under mean moderation E2 overestimates $|\beta_{bm:D}|$ by up to 46 percent (OL+BDC, $t_{1/2} = 1$) and in CO has about half the power of E1 and E3 (0.268 versus 0.538 at $c_{bm} = 0.30$), while E1 and E3 are nearly unbiased (verified). Fixing E2 is therefore a choice that favors the covariance-moderation cells.

- Parallel-RCT comparator: **resolved (2026-05-07).** Not included. S13 is concerned exclusively with the within-participant design space (cycle structure, period lengths, run-in/run-out). The parallel-RCT versus N-of-1 contrast is the responsibility of the broader nof1-design-sensitivity manuscript and is not re-evaluated here.

All open questions for S13 are now resolved. The implementation plan in §7 can proceed without further pre-implementation decisions.

References

Metadata were checked against PubMed on 2026-10-07.

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2. Hoskins M, Pearce J, Bethell A, et al. Pharmacotherapy for post-traumatic stress disorder: systematic review and meta-analysis. *British Journal of Psychiatry*. 2015;206(2):93-100.
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